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Earth and Planetary Science Letters 224 (2004) 387–398

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Combined paleomagnetic and isotopic data from the Doushantuo carbonates, South China: implications for the "snowball Earth" hypothesis

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Received 10 December 2003; received in revised form 4 May 2004; accepted 11 May 2004

Abstract

We investigated the paleomagnetism and stable isotopic signatures of the Late Neoproterozoic Nantuo tillite and overlying Doushantuo carbonates (Hunan Province, South China Block). The carbon isotope profile is typical of post-glacial series commonly observed elsewhere for the upper part of the Neoproterozoic. Together with recent radiometric dates of the carbonates, a Marinoan (ca. 600 Ma) age for the Nantuo tillite, rather than a Sturtian (ca. 750 Ma) age, is preferred. The Nantuo glacial deposits and its associated cap carbonates are remagnetized. On the contrary, a positive fold test and similarities between the Australia and South China apparent polar wander paths suggest a primary remanence for the Doushantuo Formation, which yields an equatorial paleolatitude of $3 \pm 4.5°$N. Four other paleomagnetic studies have found glacial sediments to be situated at low latitudes during this period. A distributed repartition of glacial deposits from low to high latitudes is shown on a new ca. 600 Ma paleogeographic reconstruction, strongly supporting the existence of a global Neoproterozoic glaciation.
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Keywords: South China Block; paleomagnetism; paleogeography; isotopic data; Neoproterozoic glaciation

1. Introduction

The existence of a "snowball Earth" climatic episode [1–3] implies a frozen Earth where ice or snow cover the globe, including at low latitudes. Reliable radiometric dating and good paleomagnetic determinations are essential to test the synchronicity and distribution of such glacial deposits. Despite recent progress, the existence of low latitude glacial deposits relies upon a restricted number of paleomagnetic data, such as from the Elatina Formation in Australia [4,5]. Other paleomagnetic data that give equatorial latitudes for glacial deposits are of insufficient quality in terms of paleomagnetic and/or age determinations (for a review, see [6]). Moreover, paleogeographic reconstructions between 750 and 600 Ma, often primarily based on geological and geochemical arguments, remain poorly constrained, and are complicated by suspect, rapid true polar wander (TPW) events [7].

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0012-821X/$ - see front matter © 2004 Elsevier B.V. All rights reserved.
doi:10.1016/j.epsl.2004.05.015

The Nantuo tillite and Doushantuo carbonates (Hunan, South China Block) are a good target to study such a glacial event because Lu–Hf dating and conventional Pb–Pb geochronometry independently yield well-constrained ages for the carbonates at 584 ± 26 and 599.3 ± 4.2 Ma, respectively [8]. This allows one to directly compare the paleogeography and geochemical profiles of the South China rocks with other coeval deposits, such as the Elatina Formation in Australia [4,5]. In this paper, we present a paleomagnetic and carbon isotopic study of the Nantuo tillite, post-glacial cap carbonates, and Doushantuo carbonates from the Shimen (Yangjiaping) section.

2. Geological setting

The Shimen section in Hunan Province (Fig. 1) comprises a well-preserved sequence of Neoproterozoic sediments that was folded in the Jurassic (Hunan Bureau of Geology and Mineral Resources). The Neoproterozoic sediments unconformably overlie the mid-Proterozoic Banxi group. From the base to the top are the Liantuo (also called Dieshuihe) sandstones, the Nantuo tillite, and the Doushantuo carbonates, which grade upward into the Ediacarian fauna-bearing Dengying Formation [9] (Fig. 2A and B). Upper and lower boundaries of the glacial deposits are mainly erosional. A layer of thinly laminated carbonates (4 m thick) at the base of the Doushantuo Formation is found elsewhere on the Yangtze platform and likely represents post-glacial cap carbonates [10]. The section is then comprised of 276 m of carbonaceous shales with intercalated carbonates, often poorly exposed, which are in turn overlain by 270 m of limestone and dolomite, intercalated with phosphate-rich horizons. Sedimentological structures like spherical oolithes are well preserved.

3. Carbon isotopes and petrographic observations

We measured the carbon and oxygen isotopic compositions of CO_2 of 85 specimens separated from 56 samples. Between 1 and 10 mg of powdered sample was reacted with H_3PO_4 at 25 °C for 4 h to extract the

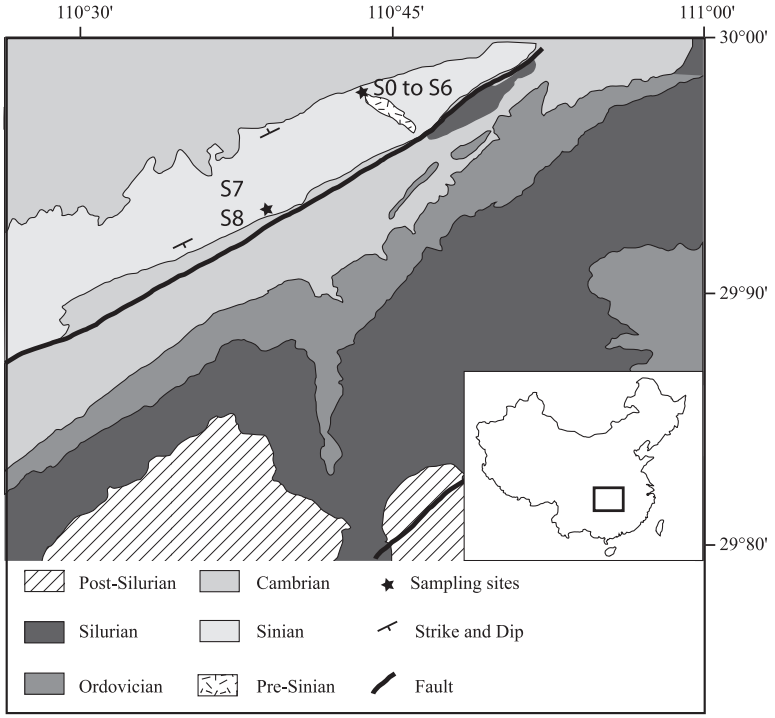


Fig. 1. Geological map of the Shimen section, Hunan province.

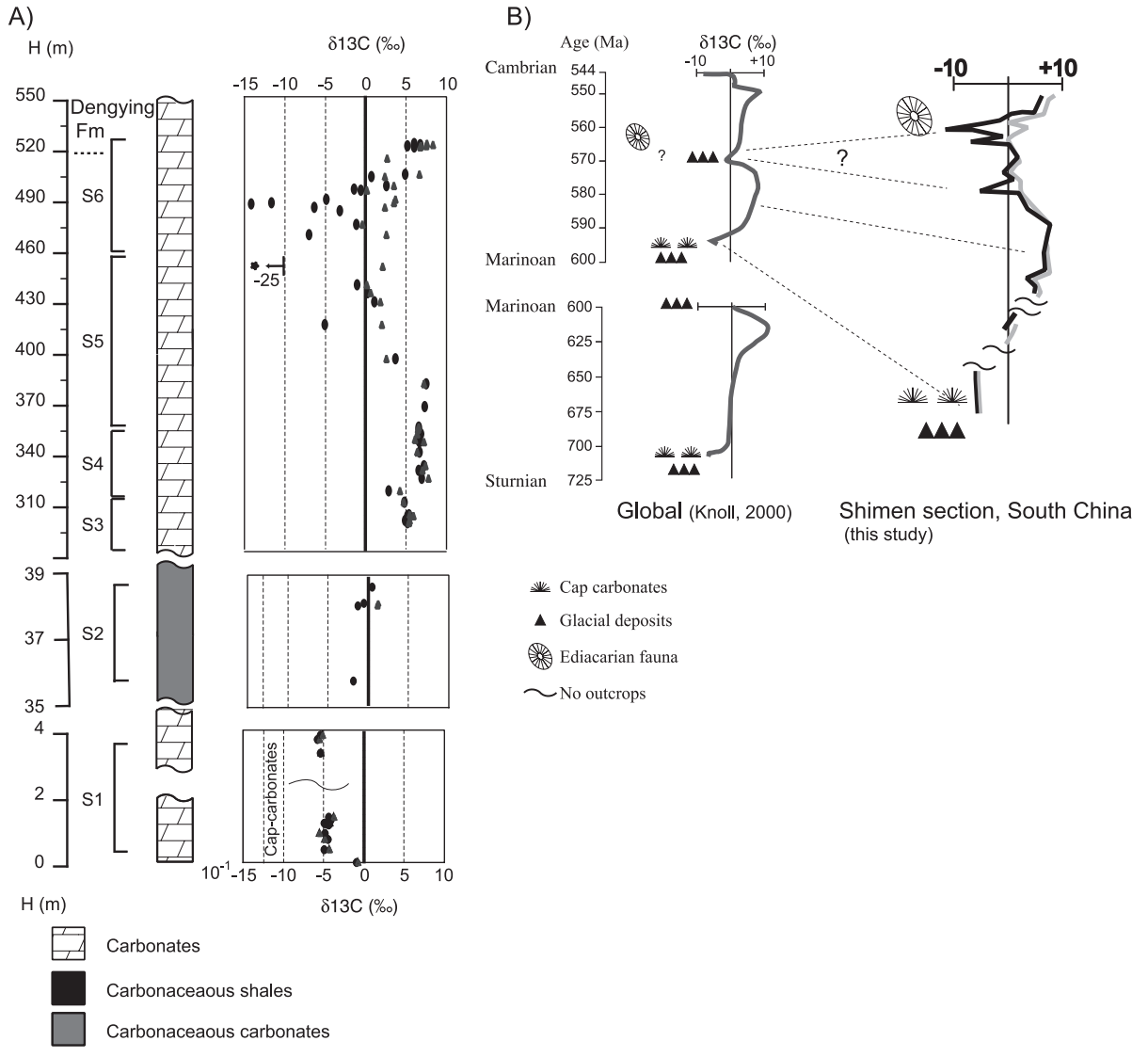


Fig. 2. Isotopic data. (A) Stratigraphic variations of $\delta^{13}C$ calcite (black dots) and dolomite (grey triangles). Thicknesses are in m above an arbitrary baseline. (B) Correlation between the composite C-isotope chemiostratigraphy of the late Neoproterozoic modified from [12] and the Shimen section (black and grey lines are respectively calcite and dolomite $\delta^{13}C$). The exact timing of earlier events is conjectural. Triangles indicate ice ages.

CO_2 from the calcite and then at 80 °C for 2 h to extract the CO_2 from the dolomite [11]. The amount of extracted CO_2 and its carbon and oxygen isotopic compositions were measured using a helium continuous-flow mass spectrometer (AP-2003). The reproducibility of the $\delta^{13}C$ measurements is $\pm 0.1‰$. Data are summarized in Table 1 and in Fig. 2.

The first meter of thick laminated limestone (Fig. 2A) is characterized by strongly negative $\delta^{13}C$ values ($-4‰$ to $-6‰$), which is typical of cap carbonates [1]. Between 35 and 39 m above the tillite, calcite and dolomite $\delta^{13}C$ values are 0‰. This trend toward positive values appears to continue up section, with $\delta^{13}C$ values of $+5‰$ to $+7‰$ from 300 to 385 m

Table 1
Isotopic results from the Doushantuo Formation

Sample	H (m)	N	$\delta^{13}C$ (cal.) (‰)	$\delta^{13}C$ (dol.) (‰)
do1002	0.4	2	−4.8	−4.3
do1004	0.7	2	−4.5	−5
do1006	0.88	2	−4.9	−5.5
do1008	1.13	2	−4.4	−4.5
do1010	1.18	2	−4.8	−4.2
do1011	1.28	2	−4.2	−4.3
do1012	1.36	2	−4.4	−3.8
do1013	3.3	2	−5.4	−5.4
do1014	3.71	2	−5.8	−5.6
do1015	3.84	2	−5.5	−5.2
93	10.48	2	–	–
94	10.68	2	–	–
96	11.07	4	–	–
98	11.47	2	–	–
100	35.8	4	−1.9	
104	38.07	4	−1.3	1.2
105	38.14	4	−0.5	1.3
107	38.64	4	0.4	2
109	295.57	4	5.2	5.4
113	296.96	4	4.9	5.3
115	299.24	4	5.2	5.9
do0501	300.53	2	5.4	5.5
do0506	307.48	2	4.8	4.7
do0510	314.32	2	2.8	4.3
do0603	321.58	2	7	7.7
119	326.5	4	6.6	7.1
120	329.5	4	7.2	7.3
122	337.28	4	6.7	6.4
do0615	343.07	2	6.7	6.9
124	344.87	4	6.5	6.2
125	347.87	4	6.9	6.4
126	350.63	4	6.6	6.6
128	352.32	4	6.5	6.5
131	364.46	4	7.3	–
135	377.85	4	7.5	7.1
do0305	392.77	2	3.7	2.5
do0311	412.37	2	−5.1	2
do0319	426.04	2	1.1	1.8
do0323	431.82	2	0.2	0.6
do0327	436.25	2	−1.1	0.2
do0329	447.08	2		2
do0343	465.65	2	−7.2	2.5
do0347	471.94	2	−1.2	−0.4
do0353	479.94	2	−3.2	–
174	484.04	2		–
175	484.86	4	−11.7	3.6
177	486.49	4	−4.9	3.6
181	491.83	4	−0.5	0.2
182	492.56	2	−1.5	–
183	494.64	4	2.6	3.5
185	499.94	6	0.7	2.4
186	501.43	4	4.9	6.6

Table 1 (*continued*)

Sample	H (m)	N	$\delta^{13}C$ (cal.) (‰)	$\delta^{13}C$ (dol.) (‰)
do1101	502.16	2	−22.7	−22.5
do1105	511.03	2	−20.9	2.7
192	517.91	4	6	6.8
193	518.12	4	6	7.5
172	518.33	4	−6.4	2.5
194	518.33	4	5.2	6.9
197	518.73	4	6.8	8.3
do1109	520	2	6.1	7.6

Column headings indicate the following: sample number (sample), stratigraphic position above the top of the Nantuo formation in m (H), number of measured subsamples (N), $\delta^{13}C$ from calcite (cal.) in ‰ and $\delta^{13}C$ from dolomite (dol.) in ‰.

above the tillite. Higher up (above 385 m), the dolomite $\delta^{13}C$ values drop to between 0 and +3‰. The high carbonate content (>50%), the lack of recrystallization observed in thin section and the agreement between calcite and dolomite $\delta^{13}C$ values support a primary nature of the C-isotopic signal in the rocks from 0 to 450 m.

There is a notable drop in calcite $\delta^{13}C$ to values between −3‰ and −21‰ at around 490 m. Observation at the outcrop revealed several fractures filled with white carbonate in the dark grey matrix of the carbonate, and a peculiar sedimentary structure resembling bubbles (around 1 cm in diametre) of white carbonate. Thin sections show recrystallizations and sparitic cements that fill voids and fractures. Calcite $\delta^{13}C$ of one of the bubbles and one of the fractures are between −22.7‰ and −20.9‰, suggesting that secondary carbonate crystallized out of fluids enriched in ^{12}C before lithification of the sediment. The strong and irregular difference between calcite and dolomite $\delta^{13}C$ probably results from variations in the amount of secondary carbonate in these samples. Such a process should lead one to interpret C-isotopic negative shifts with great caution, as they may represent local diagenetic processes rather than global worldwide changes in the carbon cycle. On the other hand, this shift to negative values is found worldwide around 570 Ma [12] (Fig. 2B). Finally, at the top of the section, $\delta^{13}C$ of dolomite and calcite return to more positive values of +7 ± 1‰ and +6 ± 1‰, respectively. Similar variations in $\delta^{13}C$ have also been found in other sections from the Yangtze platform (see a review in [13]) and, for this reason, we favor a global origin to explain the

C isotopic variations. If true, this would mean that the factors controlling diagenesis were likely attributed to a worldwide event.

The presence of cap carbonates at the base of the Doushantuo Formation, which overlie the Nantuo tillite, the presence of Ediacaran fauna in the overlying Dengying Formation, and the correlation in stratigraphic variations in $\delta^{13}C$ isotopes with Marinoan-aged sections [12], support a Marinoan age (ca. 600 Ma) of glaciation for the Nantuo tillite (Fig. 2B) (see also [13]). This is consistent with the ca. 600 Ma age from Lu–Hf and Pb–Pb dating [8] and argues against a Sturtian age (ca. 750 Ma) as previously thought (see [14]). Moreover, the Nantuo tillite cannot be correlated with a glacial epoch identified to be younger than 600 Ma [12].

3.1. Paleomagnetic results

Seven sites (185 samples) from the Nantuo and Doushantuo (S0 to S6) formations were collected along the Shimen section on the road near the Dieshuihe river. Three Doushantuo sites (S7 to S9) were also sampled on the southern flank of the anticline in order to perform a fold test. In all cases, sampling was carried out using a portable gasoline powered drill. The cores were oriented with a magnetic compass and with a sun compass whenever possible. Samples were measured using 2G cryogenic magnetometers at IPGP (France), at the Geomechanics Institute of Beijing (China) and at the UCSC (Santa Cruz, USA) laboratories. All samples were subjected to progressive thermal and/or alternating field demagnetization.

Tillite samples were collected at regularly spaced intervals of ~ 1 m along its 90 m thickness. After removal of a random direction at 150 °C, a northeast and downtrending component is defined between 150 and 460–550 °C (Fig. 3a). Unfortunately, this direction lies very close to the direction attributed to a Cretaceous overprint commonly found in South China [15]. Sometimes a direction indistinguishable from the present field (PEF) value was also found. We thus think that this formation has been totally remagnetized.

Natural remanent magnetization (NRM) intensities of the carbonate samples vary from 1×10^{-5} to 1×10^{-2} A/m^2. Some zones in the Doushantuo Formation are totally remagnetized in a direction close to the PEF or the Cretaceous overprint direction. This is also the case for the cap carbonates (Fig. 3b; Table 2), some of the black slates and for some samples in site S6. Moreover, some samples in site S6 did not possess stable magnetizations. In the other levels, these "recent" overprints are isolated at the low unblocking temperatures between 150 and 350–380 °C (Fig. 3c and d). The low temperature component in samples from the southern flank of the fold has lower inclinations than those found in the Shimen section. At higher temperatures (400–550 °C), samples from both flanks display a second component that decays linearly to the origin. Given the unblocking characteristics, the remanent magnetizations are likely carried by titanomagnetite (Fig. 3c and d). Often, above 475–500 °C, magnetic susceptibility increases markedly and the directions become unstable, indicating a change in magnetic mineralogy occurred during heating. In these cases, unblocking trajectories often define great circle swaths without arriving at a stable end point (Fig. 3e and f). We fit great circles to these samples and combined them with linear data, which are all solely of normal polarity, to calculate site mean directions (Table 2) [16]. An overall mean direction of the seven site means lies at $D = 91.0°$ and $I = 6.6°$ ($k = 45.0$; $\alpha_{95} = 9.1°$) after tilt correction (Fig. 3g and h). The McElhinny [17] fold test is positive at the 95% confidence level, and the McFadden [18] fold test is positive at 99% for the Xi1 parameter. Thus, assuming that the Earth's magnetic field was dipolar and geocentric (see discussion in [19]), the mean pole (Lat = 0.6°, Long = 196.9°, $d_p = 4.5°$, $d_m = 9°$) gives an equatorial paleolatitude of $3.0 \pm 4.5°$ for the Doushantuo Formation.

4. Discussion

4.1. Correlation between C-isotope and magnetic signal: demagnetization processes

Fig. 4 shows $\delta^{13}C$ plotted as a function of NRM intensity. Two different trends are apparent. The first one has samples with positive $\delta^{13}C$ (>5‰) values that are invariant with respect to NRM intensities. These same samples possess the high temperature component. The second trend exhibits a correlation between $\delta^{13}C$ <5‰ and NRM intensity. Samples lying on this trend are associated with remagnetization, have the weakest NRMs or have unstable magnetizations (such

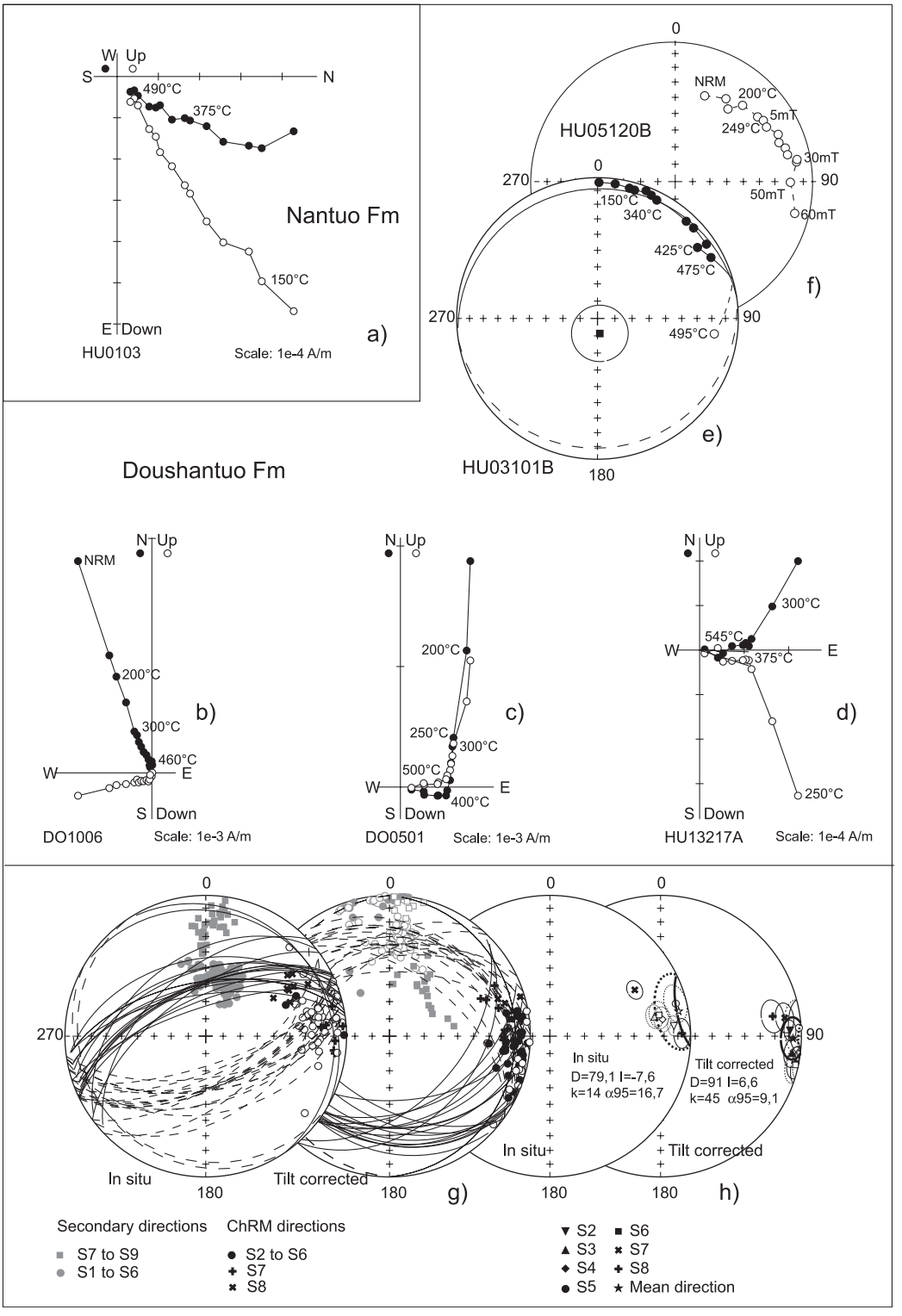

W Up
S
N
490°C
375°C
Nantuo Fm
150°C
E Down
HU0103
Scale: 1e-4 A/m
a)
0
NRM
200°C
5mT
249°C
30mT
HU05120B
270
0
90
50mT
150°C
60mT
340°C
425°C
475°C
f)
270
90
495°C
e)
HU03101B
180
Doushantuo Fm
N Up
NRM
200°C
300°C
460°C
b)
W
E
S Down
DO1006
Scale: 1e-3 A/m
N Up
200°C
250°C
c)
300°C
500°C
W
E
400°C
S Down
DO0501
Scale: 1e-3 A/m
N Up
300°C
545°C
W
E
375°C
d)
250°C
S Down
HU13217A
Scale: 1e-4 A/m
0
0
0
0
270
90
In situ
D=79,1 I=-7,6
k=14 α95=16,7
Tilt corrected
D=91 I=6,6
k=45 α95=9,1
In situ
Tilt corrected
In situ
180
Tilt corrected
180
180
180
g)
h)
Secondary directions
S7 to S9
S1 to S6
ChRM directions
S2 to S6
S7
S8
S2
S3
S4
S5
S6
S7
S8
Mean direction

Table 2
Site mean ChRM direction and mean direction for all sites

Site mean directions							
Site number	n	D	I	k	α_{95}	D*	I*
S2	4	86.4	−14.5	721.0	3.6	89.7	8.8
S3	14	79.4	−25.4	42.0	6.2	97.3	6.4
S4	17	81.5	−20.2	29.2	6.7	101.2	2.4
S5	8	75.5	−8.5	25.6	11.6	86.5	−1.8
S6	16	79.0	−22.2	23.2	7.9	102.0	−2.3
S7	10	61.4	31.9	52.5	6.7	81.6	11.8
S8	11	88.8	6.7	35.2	7.9	80.0	20.2

Mean direction for all sites

	N	D	I	k	α_{95}	D*	I*	K*	α_{95}*
	7	79.1	−7.6	14.0	16.7	91.0	6.6	45.0	9.1

Column headings indicate the following: number of samples (n), number of sites (N), in situ mean direction (D) and inclination (I), the best estimate of Fisher's precision parameter (k), radius of the 95% circle of confidence for the mean direction (α_{95}). Tilt corrected direction (D*) and inclination (I*) with their K* and α_{95}*. Site coordinates S2 to S3: N29°58′ 20″, E110°43′ 35″; S7 and S8: N29°55′ 15″, E110 °42′ 58″.

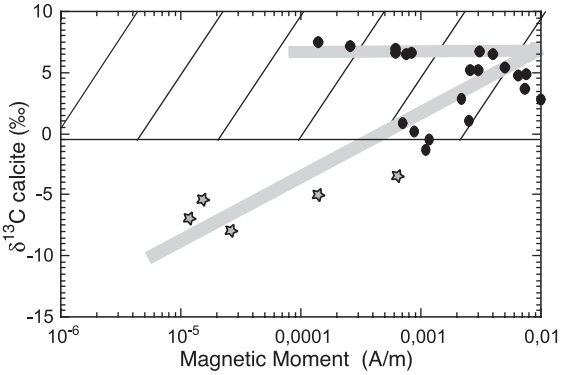


Fig. 4. Magnetic moment plotted as a function of $\delta^{13}C_{calcite}$. Solids dots are samples that provide the ChRM and stars are either remagnetized or unstable samples. Light grey lines highlight two main tendancies. The horizontal one reflects the behavior of samples localized at the top of the section, the Dengying formation. The other one defines the relationship between secondary carbonate crystallization and perturbation of both the isotopic and paleomagnetic signals (see text).

as those from site S6). Disturbance of both paleomagnetic and isotopic signals occurred most probably within the soft sediment, and is only localized within particular levels. This suggests the primary magnetic signal was acquired early during diagenesis.

Remagnetization processes that affect on one hand glacial sediments and cap carbonates and on the other hand carbonaceous slatey rocks seem different. Different rock compositions lead to specific modification of both the paleomagnetic and the isotopic signals [20]. Mechanisms explaining the acquisition of low δ^{13}C and a contemporaneous loss of magnetization are currently under investigation.

4.2. Correlation between the South China and Gondwana APWPs

Poles of often poor quality have been published over the last 20 years, so that the construction of a South China Block apparent polar wander path (APWP) for the Early Paleozoic and Neoproterozoic is not a simple task. In making our reconstructions, we selected poles using the GPMDB database Ver. 4.3 (2002). Only two selection criteria were retained: α_{95} lower or equal to 15°, assuring a minimum precision, and "demagcode" at least equal to 4, meaning that the data analyses used Zijderveld plots and principal component analysis, which today is a minimum requirement. A very sparse list of poles results from this selection (plotted in Fig. 5A). Note

Fig. 3. Paleomagnetic results from the Nantuo and Doushantuo formations. (a) Orthogonal vector diagrams of progressive thermal demagnetization for the Nantuo glacial formation samples. The closed/open symbols refer to the horizontal/vertical plane (in situ coordinates). (b–d) Orthogonal vector diagrams of progressive thermal demagnetization for samples of the Doushantuo carbonates (tilt corrected coordinates). The first panel (b) shows an example of a cap carbonate sample that only yielded the Mesozoic remagnetization direction often found in China [15]. The two following panels represent vector ChRM determination first with a secondary component from the NRM to around 400 °C and then the ChRM from 375 to 500 °C (c, d). (e, f) An example of a great circle demagnetization path isolated by thermal demagnetization only (e) and of thermal demagnetization (250 °C) followed by AF demagnetization (f). (g, h) Equal-area projections of secondary (light grey) and ChRM magnetic component directions (black great circles and black dots or open circles) observed at the sample level (g) or site level (h). The secondary component directions are more dispersed after tilt correction (g) while there is a better cluster of ChRM directions (h). Solid (open) circles represent directions plotted onto the lower (upper) hemisphere.

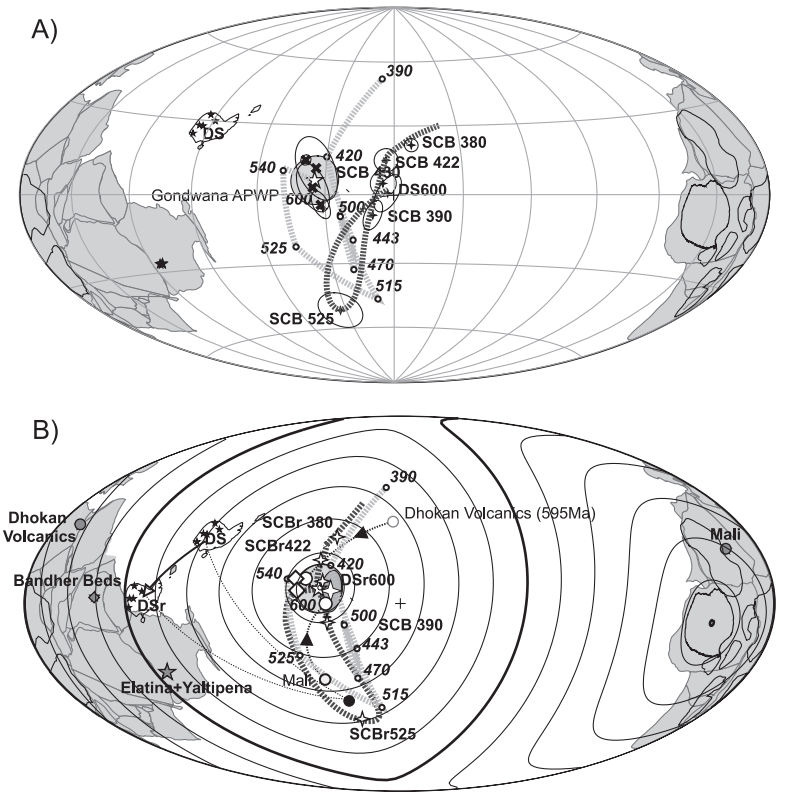


Fig. 5. (A) Gondwana [25] and South China (see text) apparent polar wander paths APWP between 380 and 600 Ma (respectively drawn as light and dark grey hatched lines). South China and associated paleomagnetic poles are shown in present-day coordinates. Gondwana reconstruction is in fixed South African present coordinates (see reconstruction parameters in [35]). Small black stars: site localization for south China and Australia. Small black four-pointed stars: selected South China block (SCB) paleomagnetic poles with their ages and uncertainties (see text). DS600 is the ca. 600 Ma pole from Doushantuo carbonates (this study). Small open dots: mean poles for Australia by [25]. The black crosses and the large open star represent the Marinoan poles derived from the Elatina and Brachina formation in Australia, respectively [5]. (B) Reconstruction of the South China Block with respect to Gondwana at 600 Ma using their APWPs (respectively figured in light and dark grey hatched lines). The South China Block and its APWP are rotated with respect to South African blocks using an Euler pole at λ=43°S, ϕ=162°E, angle=36° ccw (black dot). Notice the good agreement between both APWPs; note also the closeness between Late Neoproterozoic and Silurian poles for both China and Australia. The Indian continent is placed at low latitude using the pole (diamonds) of [32] redated by [33,34] as discussed in the text. This position is compatible with the Dokhan volcanics pole [36] from the Arabian–Nubian shield and with the pole of Morel [38], from Mali, assuming local rotations about a vertical axis (see the small circle in dotted grey line centered on the site and passing through the pole). Grey and white symbols represent, respectively, site locations and associated paleomagnetic poles.

that discarded poles, which appear in pole lists of other publications, suffer from other important defaults. Our list comprises a Devonian pole by Fang et al. [21], two Silurian poles of Opdyke et al. [22] and Huang et al. [23], and a Cambrian pole by Yang et al. [24]. At face value, these poles define a long southward track between 420 and 520, a cusp at 520, then a northward track extending to the 600 Ma Doushantuo pole (Fig. 5A).

Because our Doushantuo pole (Fig. 5A) lies close to the Early Silurian pole and because the positive fold test only implies a pre-Jurassic age of folding, the Doushantuo carbonates could have been remagnetized in the Early Silurian. Similarly, the Devonian pole of Fang et al. [21] lies close to some pre-Devonian poles from South China, which led them to suggest that a Devonian remagnetization affected the older rocks. This was contradicted by Huang et al. [23] who

suggested that Carboniferous to Cambrian poles define a fairly constant track, with the exception of the Devonian pole [21] that they omitted from consideration due to a small number of samples and lack of field test. An alternative explanation is that the site corresponding to the Devonian pole has since undergone a clockwise rotation. Undoing this rotation would place the pole on the 390 Ma part of the South China track [23]. The resulting South China APWP would then be remarkably similar to the Gondwana APWP (Fig. 5A) [25], which is defined by a long, continuous track from 390 to 515 Ma, a cusp at 515 Ma, then another track from 515 to 540 Ma. On the Gondwana APWP, one observes that the 420 and ~ 600 Ma poles (which comes from the Elatina series that is nearly coeval with Doushantuo) lie close to one another. Furthermore, the Gondwana APWP presents the same features in shape, time and amplitude as the South China APWP, such as a similarity of the Early Silurian and Late Neoproterozoic poles and the loop between the Early Vendian and Silurian. We think therefore that proximity between the 600 and 420 Ma poles is a robust feature of the South China APWP.

Another problem is that the Doushantuo carbonates have only a single polarity (normal). This contrasts with the Marinoan deposits in Australia that contain several reversals [5]. However, the Australian results come directly from the tillite layers, whereas Doushantuo lies above the tillite and cap carbonates and is thus younger. Moreover, the samples possessing primary remanent magnetizations are located between 280 and 520 m above the tillite, for a thickness of 240 m. Typical sedimentation rates in such environments are estimated between 5 and 100 cm/ky [26], corresponding to a duration between 240 ky and 4.8 Ma, assuming constant sedimentation, which is compatible with established magnetic polarity durations [27]. We think that the Doushantuo magnetization is likely of primary origin based on the fact that (1) the isotopic record was acquired early during diagenesis, (2) no deformation affected the section until the Mesozoic and (3) the similarity between the South China and Gondwanaland APWPs (Fig. 5A) [25].

4.3. Reconstruction at ca. 600 Ma

Similarities between the South China and Australia APWPs suggest that the two once formed a larger plate, as first suggested by Yang et al. [24] on the comparison of Silurian and Cambrian poles. Moreover, by adding the 600 Ma poles to the respective APWPs, we find a unique Euler pole ($\lambda = 43°$S, $\phi = 162°$E, angle $= 36°$ counterclockwise; in South African coordinates, and $\lambda = 6.3°$, $\phi = 147°$, angle $= 68.8°$ counterclockwise, in Australian coordinates) that allows for an improved fit between the two (Fig. 5B). Indeed, a reconstruction at ca. 600 Ma using the Marinoan glacial deposit poles of South China and Australia implies that South China was situated near the western part of Australia (Figs. 5B and 6). Such a reconstruction is supported by similarities in Neoproterozoic and Paleozoic geology between the two plates, such as Cambrian trilobites, which have long been recognized [28]. A previous Neoproterozoic paleogeographic reconstruction that suggested a divergence between Australia and South China [29] (Fig. 3c) was based mainly on the fact that Marinoan glacial deposits were previously unrecognized in South China; thus, this previous reconstruction is no longer supported by the emerging paleomagnetic, radiometric and isotopic data. From the APWPs, the South China and Australian blocks were together at least through the Silurian [30].

A paleogeographic reconstruction at 600 Ma is not a simple task, largely due to the lack of reliable paleomagnetic data for some of the blocks. Moreover, the final amalgamation of the Gondwana Supercontinent was probably not achieved before the end of Precambrian, at around 540 Ma [31]. In our 600 Ma reconstruction (Fig. 6), we first reconstructed the east-central part of Gondwana by using the Bandher group pole from India [32]. Within the Bandher group, De [33] identified Ediacara fauna in the Lakhery Formation, below which the isotopic stratigraphy within the formation most probably includes Marinoan-aged strata [34]. The Bandher group pole can thus be compared to the Elatina Formation pole, from which it is indistinct at the 95% level using the classical Gondwana fit [35]. India and Australia thus lie in close proximity (Fig 6), following [29]. This differs from the 580 Ma reconstruction proposed by Meert [31] where India and Australia are separated by an ocean, ‘‘the Mawson Sea’’. India and the eastern part of African blocks, such as the Arabian–Nubian shield, were probably the same plate at this period, albeit some deformation may have occurred between them [31]. This can be tested by using the Dokhan volcanics pole [36] from Egypt,

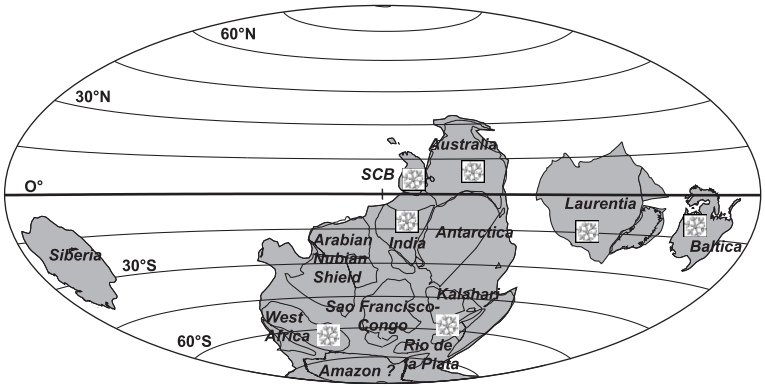


Fig. 6. Possible paleogeographic reconstruction at ca. 600 Ma. Positions of Australia, South China and other blocks are discussed in the text. Snowflakes represent the position of Marinoan glacial deposits (ca. 600 Ma). Snowflakes in black boxes have associated paleomagnetic data to constrain their locations.

which was recently re-dated at 593 ± 15 Ma [37]. By assuming a local vertical axis counterclockwise rotation of 48.5°, this pole coincides well with the bulk of Indian Bandher beds of the same age (Fig. 5B). Another pole by Morel [38] from Mali has been radiometrically dated at 616 ± 11 Ma. This pole, computed from normal samples only in the database has in fact a field test based on antipodal normal and reversed samples slightly younger in a nearby equivalent formation. It also falls on the Australian + South China APWP (Fig. 5B) assuming a counterclockwise rotation of some 90°. There is no reliable 600 Ma pole for the Sao Francisco-Congo, Kalahari, Rio de la Plata, and West Africa plates. However, based on petrologic and geochemical arguments, Alkmim et al. [39] and Meert [31] suggest that these blocks were in contact. We follow [31] when placing these blocks, which is in contrast to recently published paleomagnetic data from Brazil [40]. Although the Brazilian data have not passed a fold test, the directions are antipodal. However, the mean direction is indistinguishable with the recent geocentric axial dipole field. Further paleomagnetic data with a stronger field test from the region are required.

A 608 ± 1 Ma pole by Eneroth and Svenningsen [41] provides a robust paleolatitude for Baltica, whereas the paleoposition of Laurentia remains controversial. Many of the arguments surround the 577 Ma Catoctin metabasalts (Canada), which possess two magnetic components—one with a high paleolatitude and another with a low paleolatitude [42]. Another Laurentian paleomagnetic pole from the Callander complex (Canada) [43] also indicates a high paleolatitude. Other poles at 615 and 565 Ma suggest Laurentia was situated at equatorial latitudes [44, Fig. 9]. If one tries to resolve all the data, it would suggest that Laurentia moved 60° in latitude from 615 to 577 Ma, then returned to an equatorial setting, moving 70° north from 577 to 565 Ma. The latter motion implies an average plate velocity of 64 cm/year, which is incompatible with modern plate speeds. Moreover, during the same time interval, only limited motion is implied from the Australia and India APWPs, which argues against a rapid true polar wander event. Thus, following Hodych et al. [44], we favor a low latitude position for Laurentia at ca. 600 Ma. Finally, the position of Siberia was derived from the pole list of Smerhurst et al. [45].

Our reconstruction suggests the globe was dominated by a large continent (Gondwana) that spanned from the equator to the South Pole, with a few large continents interspersed at tropical to equatorial latitudes (Fig. 6). This reconstruction should be considered tentative. Given the uncertainties on ages and paleomagnetic poles, our version clearly needs to be substantiated by further work in several areas of the globe, especially concerning the West African and South American blocks. With this caveat in mind, we placed the known Marinoan glacial deposits from Australia, South China and India on the map (Fig. 6). Equivalent glacial deposits have also been found in the Mackenzie Mountains of Laurentia (Ice Brook Formation) [46], in Namibia (Ghaub formation, [2])

and in West Africa (Volta Basin, [47], which were correlated with the Marinoan Australian Elatina Formation, and thus with the Marinoan Indian and Chinese glacial formations. Our reconstruction suggests that glacial deposits were distributed from low to higher latitudes and advocates the existence of a global glaciation at ca. 600 Ma (Fig. 6).

Finally, we note that if the Marinoan glaciation is dated slightly older than 600 Ma (625 Ma, P. Hoffman, personal communication, 2004), our conclusions remain valid due to the limited amount of polar wander suggested by the Australian APWP.

5. Conclusion

Although we isolated only a single polarity in the Doushantuo Formation, isotopic arguments that the remanence was locked-in at the early stages of diagenesis, together with a positive fold test and similarities in both the Australia and South China APWPs, suggest a primary origin for the paleomagnetic remanence. A well constrained age of 599 ± 4 Ma of the Doushantuo carbonates [8], similarities in stratigraphic $\delta^{13}C$ variations with those found in worldwide compilations [12], the close resemblance of the South China and Australian APWPs and the presence of Ediacarian fauna in the top of the Shimen section strongly support a Marinoan age for the Nantuo tillite. Assuming the magnetization is primary, we obtain a paleolatitude of $3.0 \pm 4.5^\circ$ for the Doushantuo carbonates, suggesting the Nantuo tillites were deposited close to the equator. Our 600 Ma reconstruction situates South China close to western Australia, which formed part of a larger Gondwanan continent. This reconstruction gives strong support for the existence of low latitude glacial sediments and supports of the existence of a global Neoproterozoic glaciation.

Acknowledgements

The authors would like to thank R. Van der Voo, Y. Godderis and P. Hoffman for their constructive reviews. These comments were of considerable assistance in shaping the published version of this article. This study was supported by PRA (French–China No. 01-01) and by Eclipse project (No. 367). We are indebted to M. Qirui Zhang for their participation in the fieldwork. We are also indebted to J.-P. Cogné for providing the excellent software PALEOMAC [48]. This is IPGP contribution No. 1998. ***[BARD]***

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